University of Geneva
Department of Informatics

Double JPEG Compression

Digital Forensics

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Michel Jean Joseph Donnet

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1 Introduction

The modification of information exists and has been practiced for a long time, both by governments and by individuals. The field of photography is particularly prone to this. This can take the form of a detail being corrected or added to improve a photo, or the addition or removal of important information, such as the presence of people in a specific photo. This can have serious repercussions, particularly in the field of justice. Proof of a person's presence in a particular place at a given time can be critical to the resolution of a case. At this point, the digital forensic enter in game and, thanks to its methods, can determine a lot of useful informations about a photo, like if the photo was modified, which kind of device take the photo etc... When someone modify a photo, he must save the changes and write the modified photo on the disk. This work will study the detection of this kind of manipulations by detecting double JPEG compression. Depending on manipulations, it's possible to determine if a photo was modified.

2 Methodology

In the digital world, the compression of informations plays a crutial role in multiple domain by reducing the size of the data while preserving the information. In imagery, the compression allow to store more photo on disk with or without (depending on method used) loss of informations.

There are many kind of compressions that exists, as `png` or `jpeg` compression. Some allow compression without data loss (lossless compression), while others perform compression with data loss (lossy compression).

In the remainder of this report, we will focus on a lossy compression: `jpeg` compression. But first, we will explain the mecanisms bellow image compression.

2.1 Image compression

The image compression is based on redundancies, both spatial and psychovisual.

- Spatial redundancies. This kind of redundancies are present in some region of image. For instance, at some given region, all pixels can have the same intensity. So it's possible to deduct intensity of given pixel by taking the neighbors of this pixel.

- Psychovisual redundancies. This is based on the imperfection and characteristics of Human Visual System. For instance, Human Visual System is more sensitive to the brightness than to the colors, so image can be transformed into YCbCr channels with Y the luminance of the image and Cb and Cr some color channel. In this way, we can conservate original luminance of the image, and downsampling the colors channels, and the Human Visual System will not see any change in the image. Human Visual System is also less sensitive to high frequencies, that's why a DCT (Discrete Cosine Transform) transform is applied, allowing to separate low and high frequencies. Then, a quantization is made, removing the high frequencies, and Human Visual System will not see the difference.

So it's possible to reduce the quantity of data to store by deleting redundancies in the image. The general way to compress an image lies in this way:

- First, the data are transformed in a more amenable way to compression, to exploit redundancies. For instance, applying a DCT allow to separate low and high frequencies.
- Then, a quantization is performed in lossy compressions. The quantization is a way to delete non useful informations of the image for Human Visual System, like high frequencies. For a lossless compression, this step is ignored to avoid a loss of informations by quantization.

- Finally, the new data obtained are encoded and stored in the device in such a way it can be decoded to visualize the image.

2.1.1 Quantization

In the quantization step in JPEG compression, a quantization matrix Q is applied for each block of DCT coefficients. Suppose that we have a matrix of coefficients C_{ij} of DCT, and a quantization matrix Q for the given channel. Then, the quantization is applied as in the formula 1.

$$C_{ij\ \text{quantized}} = round\left(\frac{C_{ij}}{Q}\right) \tag{1}$$

The quantization matrix looks like in the formula 2 and can be ajusted depending on needs.

$$Q = \begin{bmatrix} 17 & 18 & 24 & 47 & 99 & 99 & 99 & 99 \\ 18 & 21 & 26 & 66 & 99 & 99 & 99 & 99 \\ 24 & 26 & 56 & 99 & 99 & 99 & 99 & 99 \\ 47 & 66 & 99 & 99 & 99 & 99 & 99 & 99 \\ 99 & 99 & 99 & 99 & 99 & 99 & 99 & 99 \\ 99 & 99 & 99 & 99 & 99 & 99 & 99 & 99 \\ 99 & 99 & 99 & 99 & 99 & 99 & 99 & 99 \\ 99 & 99 & 99 & 99 & 99 & 99 & 99 & 99 \end{bmatrix} \tag{2}$$

In the formula 2, we can see that the high frequencies of the DCT will be deleted because the low frequencies are in top left corner and the closer we are to the lower right corner, the higher the frequencies are. In this way, we will divide high frequencies by big value 99, making the result equal to 0 because of the rounding.

3 Implementation

The code is implemented in python from an existant matlab code provided.

It works in command line in this way:

```
usage: tp3.py [-h] -i IMAGE [--QF1 QF1] [--QF2 QF2]

Double JPEG compression

options:
  -h, --help           show this help message and exit
  -i, --image IMAGE    Image to make double compression
  --QF1 QF1            Quality Factor for first compression
  --QF2 QF2            Quality Factor for second compression
```

The output printed is the histogram of the DCT coefficients.

4 Results

Blablablablabla

5 Discussion

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6 Conclusion

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